

Forest Sensor Sprint

Site the smartest place for a campus tree sensor

At a glance: Find the smartest place to install a future campus tree sensor. Plan about 80 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: tree height, microclimate, temperature, humidity, light, evidence-based site selection (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- 4 clipboards (one per team; 6 on hand)
- 4 paper clinometers (one per team; print 6 to 8 with spares)
- 4 temperature and humidity meters (one per team; 12 on hand)
- 4 soil meters (one per team; 6 on hand). The same meter also reads light, so no separate light meter is needed.
- 4 route cards (one per team; print 6 to 8 with spares)
- pencils

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

- 01 Set up one station per team with the materials above, sorted and ready.
- 02 Stage the scoring sheet and any reference values before testing begins.
- 03 Set out sunscreen and water, and confirm the approved outdoor route and the weather before students arrive.
- 04 Pre-measure and mark one standoff distance (for example, 10 m) from each tree used for the height estimate. The 12-inch rulers are too short for this.
- 05 Label the checkpoints (A, B, C ...) on the grounds to match the route card, and mark each with a sign or flag.
- 06 Load the activity slides and ready the projected timer.

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Find the smartest place to install a future campus tree sensor." Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints

0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Build or solve

0:18 - 0:44

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

Phase 5 Test and record

Teams run the standard test and log a baseline result before changing anything.

Phase 6 Redesign

Each team measures a second candidate site, predicts which is better, and compares the two with their data.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, and reset the station. Wash or wipe hands after handling soil, and wipe down the meters.

TROUBLESHOOTING

Problem: A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

Problem: No measurable result on the first test. **Fix:** Check the setup against the steps, confirm the standard test was followed, and log the baseline anyway.

Problem: Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to microclimate varies in meters.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Which measurement most influenced your choice?
- Where did two nearby spots disagree, and why?
- What would change your recommendation in winter?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Data accuracy	30
Route efficiency	20
Site recommendation evidence	30
Teamwork and field conduct	20
Total	100

CLEANUP AND SOURCES

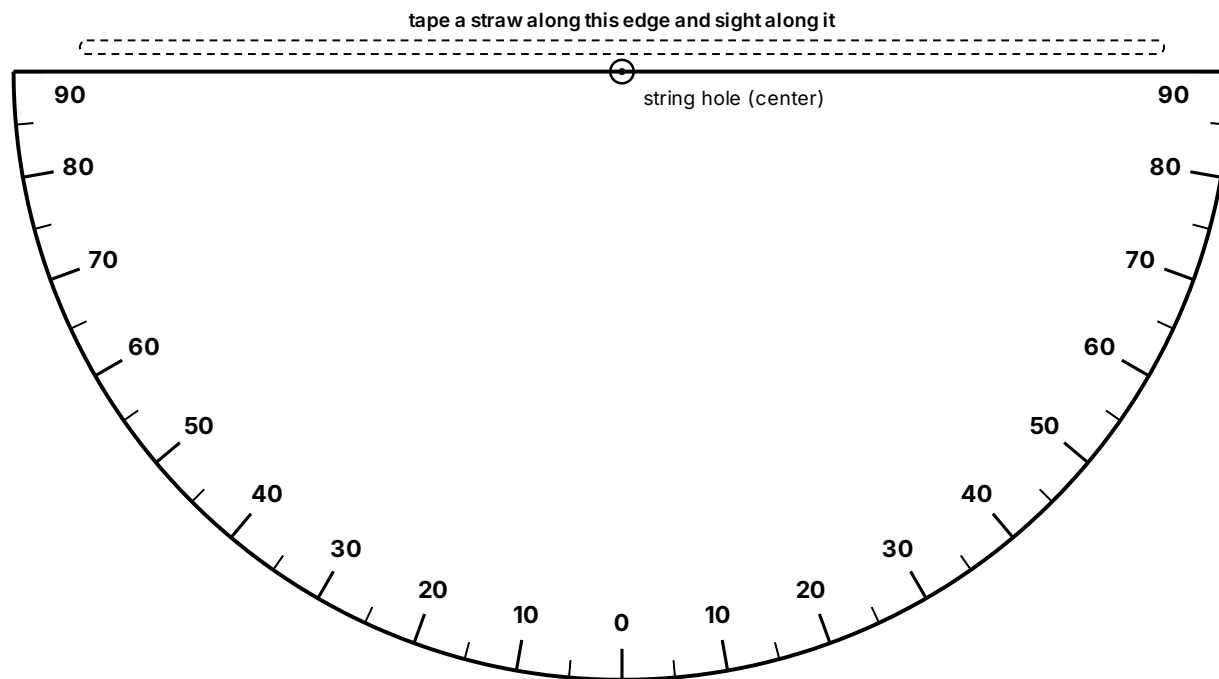
Return meters, clipboards, and clinometers to the bin, wipe down the soil meters, have students wash hands, and collect score slips.

SOURCES GLOBE Observer Trees; NASA MyNASADData Urban Heat Island

PRINT AND CUT: TEAM TOOLS

Print on cardstock for TTT-02 Forest Sensor Sprint. Clinometer: one per page, print 6 to 8. Route cards: two per page, print 3 to 4 sheets for 6 to 8 cards.

Paper clinometer



Cut along the flat top edge and the curved outline.

ASSEMBLE

- 01 Cut out the half circle along the curved outline.
- 02 Tape a straight straw along the top straight edge.
- 03 Punch the center dot, thread a string through it, and tie a washer or paper clip on the end as a weight.
- 04 Sight the treetop through the straw, let the string hang free, then pinch it against the scale and read the number.
- 05 Self-check: sight something level (reads 0) and straight up (reads 90).

FIND THE TREE HEIGHT

Tree height = eye height + (distance to the tree × tan of the angle). Use the standoff distance the instructor marked.

ANGLE	TAN
10	0.18
20	0.36
30	0.58
40	0.84
45	1.00
50	1.19
60	1.73
70	2.75

Example: eye height 1.5 m, distance 10 m, angle 40° gives $1.5 + 10 \times 0.84 = \text{about } 9.9 \text{ m}$.

Team route card

From Trees to Tech · TTT-02 Forest Sensor Sprint · Team Route Card

Team _____ Date _____

Plan an order that avoids backtracking, take a reading at each checkpoint, then recommend the best sensor site using your numbers.

CHECKPOINT	VISIT ORDER	TEMP (°F)	HUMIDITY (%)	LIGHT (LUX)	SOIL MOISTURE (1 TO 10)	NOTES
Reference (calibrate)						
A						
B						
C						
D						
E						

Our recommended sensor site: _____

Why (use your data): _____

The checkpoints are the spots the instructor marked outside.

cut here

From Trees to Tech · TTT-02 Forest Sensor Sprint · Team Route Card

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